BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions) (33)

Maximum Marks: 33

Note: Choose the correct answer for each from the given options. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 85

Q1. Choose the correct answer for each from the given options:

Qi. These elements EKa-Aluminium and EKa-Silicon were known as:

(A) Gallium and Germanium
(B) Gallium and Silicon
(C) Aluminium and Germanium
(D) Beryllium and Magnesium

Qii. Elements of group VI—A are called:

(A) alkali metals
(B) chalcogens
(C) coinage metals
(D) halogens

Qiii. Elements of this group forms an ion with charge +3:

(A) I A
(B) II A
(C) III A
(D) VA

Qiv. All elements form ionic hydrides except:

(A) Li
(B) Na
(C) Be
(D) Ca

Qv. This group forms basic hydrides:

(A) IIIA
(B) IVA
(C) VA
(D) VIA

Qvi. The half life of radioactive isotope of hydrogen is:

(A) 10.5 years
(B) 12.5 years
(C) 13.5 years
(D) 14.5 years

Qvii. This element of IA group forms a superoxide:

(A) Li
(B) Na
(C) K
(D) Sr

Qviii. This one is liberated at the anode during electrolysis of brine in diaphragm cell is:

(A) H_2
(B) Na
(C) Cl_2
(D) O_2

Qix. This one has maximum hydration energy:

(A) Be^{2+}
(B) Mg^{2+}
(C) Ca^{2+}
(D) Ba^{2+}

Qx. The oxidation state of nitrogen in HNO_3 is

(A) +3
(B) -3
(C) +5
(D) -5

Qxi. Duralumin is an alloy of:

(A) Iron
(B) Nickel
(C) Zinc
(D) Aluminium

Qxii. The basicity of ortho boric acid is:

(A) 1
(B) 2
(C) 3
(D) 4

Qxiii. All elements get corroded except

(A) Iron
(B) Copper
(C) Zinc
(D) Gold

Qxiv. The coordination number of Co in $[Co(en)_2Cl_2]^{+1}$ is:

(A) 3
(B) 4
(C) 5
(D) 6

Qxv. This metal is used for galvanizing:

(A) Fe
(B) Co
(C) Ni
(D) Zn

Qxvi. The hybridization of all carbon atoms in cyclohexane (C_6H_{12}) is:

(A) sp
(B) sp^2
(C) sp^3
(D) dsp^3

Qxvii. This catalyst is used as 'Knock inhibitor':

(A) Zeolite
(B) Tetra Ethyl Lead
(C) Platinum
(D) Raney Nickel

Qxviii. The general formula of ester is:

(A) RCOOH
(B) RNH_2
(C) RCOR
(D) RCOOR

Qxix. This compound obeys Markonikoff rule when reacting with HBr:

(A) $CH_3 - CH_3$
(B) $CH_2 = CH_2$
(C) $CH_3 - CH = CH_2$
(D) $CH_3 - CH = CH - CH_3$

Qxx. Vinyl chloride polymerizes into:

(A) PVA
(B) PVC
(C) Bakelite
(D) Perspex

Qxxi. The dehydration of alcohol gives:

(A) Alkane
(B) Alkene
(C) Alkyne
(D) Benzene

Qxxii. This compound gives SN^1, mechanism:

(A) Chlorobenzene
(B) Chloroethane
(C) Chloromethane
(D) 2-chloro2-methyl propane

Qxxiii. The most stable carbonium ion is:

(A) CH_3
(B) RCH_2
(C) R_2CH
(D) R_3C

Qxxiv. CH_3I reacted with sodium metal to form:

(A) CH_3OH
(B) C_2H_2
(C) C_2H_4
(D) C_2H_6

Qxxv. The colour of precipitate obtained in iodoform test is:

(A) Red
(B) Orange
(C) Yellow
(D) Green

Qxxvi. Ethyl acetate is present in:

(A) Pineapple
(B) Guava
(C) Orange
(D) Lemon

Qxxvii. This one does not contain carboxyl group:

(A) Acetic Acid
(B) Benzoic Acid
(C) Picric Acid
(D) Butyric Acid

Qxxviii. The animal starch is:

(A) Glycogen
(B) Amylose
(C) Cellulose
(D) Amino Acids

Qxxix. Blood cancer is caused by:

(A) Methane
(B) Ethane
(C) Butane
(D) Benzene

Qxxx. Bleeding of teeth is caused by the deficiency of this vitamin:

(A) D
(B) C
(C) B
(D) A

Qxxxi. Bakelite is a polymer of:

(A) Phenol — Formaldehyde
(B) Urea — Formaldehyde
(C) Phenol — Benzaldehyde
(D) Phenol — Urea

Qxxxii. This impart green colour to glass:

(A) Cr_2O_3
(B) CoO
(C) CuO
(D) ZnO

Qxxxiii. Fertilizer maintains this pH of the soil:

(A) 7 to 8
(B) 6 to 7
(C) 9 to 11
(D) 12 to 14

Section 'B' (Short-Answer Questions) (32)

Maximum Marks: 32

> Note: Answer any EIGHT part questions. Select FOUR part questions from Inorganic Chemistry and FOUR part questions from Organic Chemistry. All questions carry equal marks.

QINORGANIC CHEMISTRY.

Q2.

(i) Write the electronic configuration of the following elements having the atomic number Z =19, Z = 29, Z =35, Z= 42 and also identity group and period.

(ii) Give the resemblances of hydrogen with the elements of group IA and VIIA and write four points of each group.

(iii) Refer to the list of given compounds: Compound: A = Magnesite, B = Whiterite, C = Water glass, D = Lunar caustic. Write * Chemical formula of B and C * Reaction of A with dilute H_2SO_4. * Reaction for preparation of C. * Common uses of D.

(iv) Describe the construction of Hoope's electrolytic cell for obtaining pure Aluminium along with the chemical reaction involved.

(v) Write IUPAC names of any four of the following complexes: * $[Pt(NH_2)_3(H_2O)]^{+4}$ * $K_2[FeF_6]$ *$[Ni(CN)_4]^2$ * $[Co(en)_2](NO_3)_2$ *$[Ti(H_2O)_3]Cl_3$

(vi) Write any four chemical equations of the following: * Mg is put in dilute HNO_3. * Litharge is heated in excess of air. * Hydrogen sulphide reacts with bromine * Carbon monoxide reacts with chlorine gas * Chlorine react with conc NaOH

(vii) Give any four scientific reasons of the following * Alkaline earth metals are smaller in size than alkali metals. * Sulphuric acid has high boiling point * Ionization potential decreases from top to bottom in a given group of periodic table * Alkali metals are powerful reducing agents. * Transition metals form non-stoichiumestric compounds.

QORGANIC CHEMISTRY.

(viii) Define the following: *Cracking *Functional group *Zwitter ion *Isomerism

(ix) The structures of organic molecule (A and B) are shown below: A = ethene (H−C=C−H with one H below each carbon, i.e. $H_2C=CH_2$), B = ethyne/acetylene (H−C≡C−H). * Draw the hybrid orbital structure of 'B' with number of Sigma and Pi bonds * Give a chemical reaction of 'A' with dilute alkaline solution of $KMnO_4$. * Give equation for preparation of A by the dehydration of ethanol. * Write the chemical equation when 'B' is passed over red hot tube.

(x) What arc Electrophiles? Give the reaction mechanism of Friedel Craft alkylation and sulphonation Benzene.

(xi) Complete any four of the following chemical equations. * (phenol structure: benzene ring with −OH) + HNO_3 —(Conc. / Heat)→ * $CH_3 - CH(Cl) - CH_3 + Na$ —(Metal)→ (2-chloropropane structure: Cl below the middle carbon) * $HCHO + 2Cu(OH)_2 + NaOH \longrightarrow$ * $C_2H_5ONa + Cl - C(=O) - CH_3 \longrightarrow$ * $C_6H_6 + O_3 \longrightarrow$

(xii) Write IUPAC names any four of the following: * $(CH_3)_3\ C - (CH_2)_2 - CH(CH_3)Cl$ * $CH_3 - CH(OH) - CH_3 - COOH$ * $CH_3 - C(=O) - (CH_2)_2 - CHO$ * (structure of 2,4-dibromophenol: benzene ring with −OH and two −Br substituents)

(xiii) What are Organometallic compounds? Give the chemical equation for the preparation of 1°, 2nd, 3rd alcohol from CH_3MgI.

(xiv) Write a note on any one of the following: * Detergents * Glass * Biological importance of carbohydrates

Section "C" (Detailed Answer Questions) (20)

Maximum Marks: 20

> Note: Attempt any Two questions-One question from Inorganic Chemistry and other from Organic Chemistry. Both questions carry equal marks.

QINORGANIC CHEMISTRY.

Q3.

(a) Describe the industrial preparation of sulphuric acid by contact process.

(b) Complete and balance any five of the following chemical reactions: * H_3BO_3 + NaOH ⟶ * Zn + H_2SO_4 —(90%)→ * $CuSO_4.5H_2O$ —(230°C)→ * $KMnO_4$ + H_2O_2 —(Neutral Medium)→ * Fe_2O_3 + Al ⟶ * $K_2Cr_2O_7$ + H_2SO_4 —(90%)→

Q4.

(a) Describe the metallurgy of 99.99% copper from roasted pyrite ore.

(b) Describe the industrial preparation of soda ash by Ammonia Solvay process.

--OR-- State Mendeleev periodic law. Describe the disadvantages of Mendeleev periodic table and how it is modified by Moseley.

QORGANIC CHEMISTRY.

Q5.

(a) What is β − elimination reaction? Explain the mechanism of E_1 and E_2 reactions.

(b) Complete and balance any five of the following chemical reactions: * $2C_2H_5Cl$ + Ag_2O ⟶ * H − C(=O) − H + C_2H_5OH —(exces)→ * CH_3 − C(=O) − CH_3 + [O] —($K_2Cr_2O_7$ / H_2SO_4)→ * CH_3 − C(=O) − Cl + CH_3COONa ⟶ * $CH_3COOC_2H_5$ + NaOH —(10%)→ * (phenol structure: benzene ring with −OH) + Zn —(Dust)→

Q6.

(a) What is Orientation in Benzene? Explain orientation in monosubstituted Benzene. Give the names of ortho-para and meta directing groups.

OR Write the two possible explanations for benzene behaves as saturated and unsaturated compounds? Describe Kekule structure.

Q6.

(b) Write the structural formulae of any five of the following: * Aspartic Acid * Nicotinic Acid * Pyrogallol * α − Naphthol * Isobutyric Acid * Glycerol